How Can Coca-Cola Optimize Inventory and Distribution Across Different Markets?

PGDM | Big Data Analytics & DBMS

Evidence-based project dossier | Original uploaded dataset: 50,000 rows, 31 fields | Prepared 29 October 2026

Status: Python-based CSV analysis performed. MySQL server execution, DBeaver, Databricks/Delta Lake, Power BI dashboard and team-specific evidence: TO BE COMPLETED.

1. Executive summary and factual status

The supplied dataset spans 2023-01-01 to 2025-12-31, records 8 markets, 28 countries and 10 products, and totals USD 9,991,731.05 in recorded revenue and 11,472,087 demand units. All 50,000 records contain non-null values and there are zero exact duplicate records. This is a descriptive observational project; the source and whether the records describe actual Coca-Cola operations have not been confirmed.

Reported overstock flags occur in 42,598 records (85.20%), but source threshold logic has not been supplied. Stockout_Flag is always zero, Fill_Rate_Percent is uniformly 100, and demand equals units sold in all rows. These fields cannot establish service level differences between markets.

2. Business problem, objectives and boundaries

Objective: use one supplied data extract to assess regional demand, product revenue, forecast errors, excess inventory indications, warehousing, transportation modes and recorded costs. Develop transparent descriptive analytics to support operational prioritization, without claiming causal improvement or deployed optimization.

Unit of analysis: one CSV record, for which transaction/event grain is not documented. Warehouse IDs may repeat across countries; sums of stock snapshots, turnover or fill rates must not be interpreted as network-wide independent inventory movements without granularity validation.

3. Data provenance, methods and limitations

Dataset name: CocaCola_Inventory_Distribution_MySQL_Ready.csv. Provenance / original publisher / collection method: TO BE COMPLETED by the project team. Do not fabricate a public source URL. The data is treated as a provided academic sample until independently verified. SHA-256: eb4c0a011af2ab56b8fe9e82d08400cea6893a699703bffd62698db6e6d1d7dc.

Original CSV preserved byte-for-byte. In Python the Date field was parsed in memory; an analysis-only month key was derived for time aggregation, and no original records or fields were altered. Proposed MySQL table adds an auto-increment technical row_id at import only. Source data unchanged.

4. Dataset audit

Audit measure	Verified observation
Rows / original fields	50,000 / 31
Dates	2023-01-01 to 2025-12-31
Missing / exact duplicates	0 / 0
Invalid dates	0
Stockout / overstock flagged	0 / 42,598
Forecast MAE (unit/row)	18.29
Assumed inventory reconciliation mismatches	3,841
Revenue math mismatches	0

Inventory reconciliation test assumes closing stock = opening stock + production - units sold - waste. Its 3,841 mismatches must not be silently repaired; possible transfers, adjustments or undocumented definitions could invalidate the assumed equation. Values also contain some large numerical observations; potential outliers warrant

a subject-matter review, not automatic removal.

5. Field dictionary — complete 31-column inventory

Field	Type	Business meaning
Date	DATE	Record date
Market	VARCHAR	Regional market
Country	VARCHAR	Country associated with record
Warehouse_ID	VARCHAR	Warehouse label; not independently verified as a unique physical facility
Product	VARCHAR	Beverage product name
Category	VARCHAR	Beverage grouping
Sales_Channel	VARCHAR	Channel through which product sold
Promotion	VARCHAR	Promotional classification
Season	VARCHAR	Season as supplied (interpretation may vary by geography)
Holiday_Flag	INT	Source-provided holiday indicator; definition not independently validated
Opening_Inventory_Units	INT	Beginning stock in units

Field	Type	Business meaning
Production_Units	INT	Produced units
Demand_Units	INT	Recorded demand units
Forecast_Demand_Units	INT	Forecast demand in units
Units_Sold	INT	Units recorded as sold
Closing_Inventory_Units	INT	Reported end stock units
Safety_Stock_Units	INT	Supplied safety stock threshold; formula unknown
Reorder_Point_Units	INT	Supplied reorder trigger; formula unknown
Stockout_Flag	INT	Supplied stock-out event flag; all zeros
Overstock_Flag	INT	Supplied overstock flag; definition unknown
Lead_Time_Days	INT	Recorded delivery lead time in days
Transport_Mode	VARCHAR	Recorded transportation mode

Field	Type	Business meaning
Distance_KM	INT	Transportation distance in kilometers
Transport_Cost_USD	DECIMAL	Recorded transportation cost (USD)
Unit_Price_USD	DECIMAL	Unit selling price (USD)
Sales_Revenue_USD	DECIMAL	Recorded sales value (USD)

Field	Type	Business meaning
Inventory_Holding_Cost_USD	DECIMAL	Reported holding cost (USD)
Waste_Units	INT	Reported waste in units
Waste_Cost_USD	DECIMAL	Reported waste cost (USD)
Fill_Rate_Percent	DECIMAL	Source-provided service fill rate; constant at 100
Inventory_Turnover	DECIMAL	Source-provided turnover metric; scope/formula unverified

6. Workflow architecture and platform responsibilities

Original CSV (immutable) -> Python profiling and actual analytical charts -> proposed MySQL table -> DBeaver queries and screenshots (pending) -> Databricks Spark aggregation and Delta write (pending) -> Power BI KPI/visual validation (pending) -> final report and slides -> GitHub. Outputs must reconcile by date scope, filter, aggregation and unit definitions. SQL scripts and Spark notebook included as proposed executable artifacts, not proof of deployment.

7. MySQL and DBeaver implementation

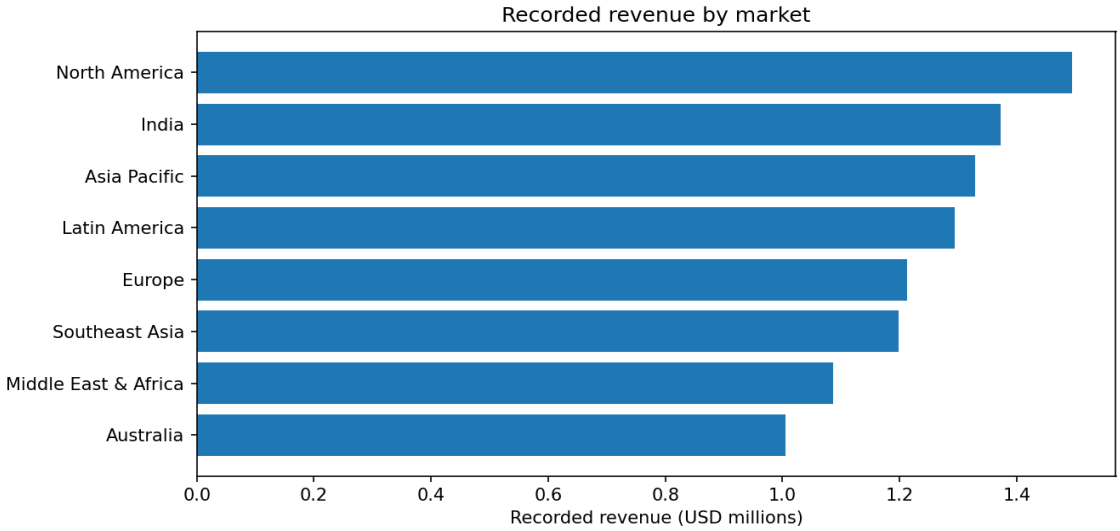
File sql/01_mysql_schema.sql contains a MySQL 8+ DDL with the 31 original fields plus a technical import ID. The flat table is retained because a verified grain and stable dimension keys are unavailable. File sql/02_mysql_analysis.sql contains 16 annotated analytic queries, including joins, CASE, subquery, CTE, window, date aggregations and validation checks. Database name: coca_cola_supply_chain (proposed).

TO BE COMPLETED: create a real MySQL connection in DBeaver, execute schema/import, capture affected row count (expect 50,000), inspect date/decimal precision, run all queries, retain screenshots and compare with supplied audit summaries. MySQL has not been accessed in this analysis environment.

8. Python analysis — executed evidence

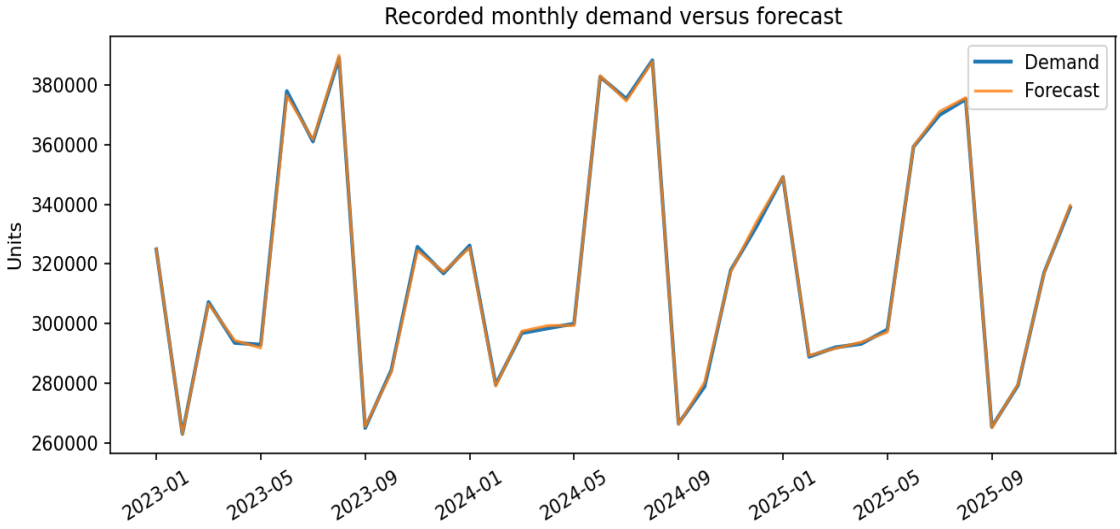
A pandas-based audit and aggregation ran against the uploaded CSV. The repository includes a fully rerunnable Jupyter notebook and four charts produced from those same source rows. Revenue is recorded USD sales rather than verified net income; transport and holding costs should not be deducted to claim profit without comprehensive cost accounting.

Figure 1. Revenue by market



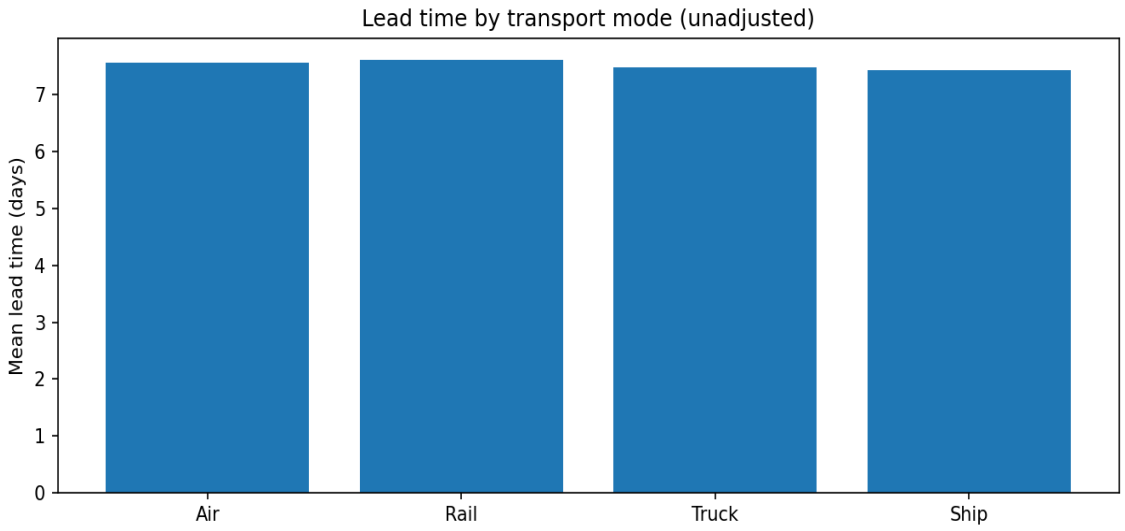
Record-level revenue summed by market; highlights concentration, not overall profitability.

Figure 2. Monthly demand and forecast



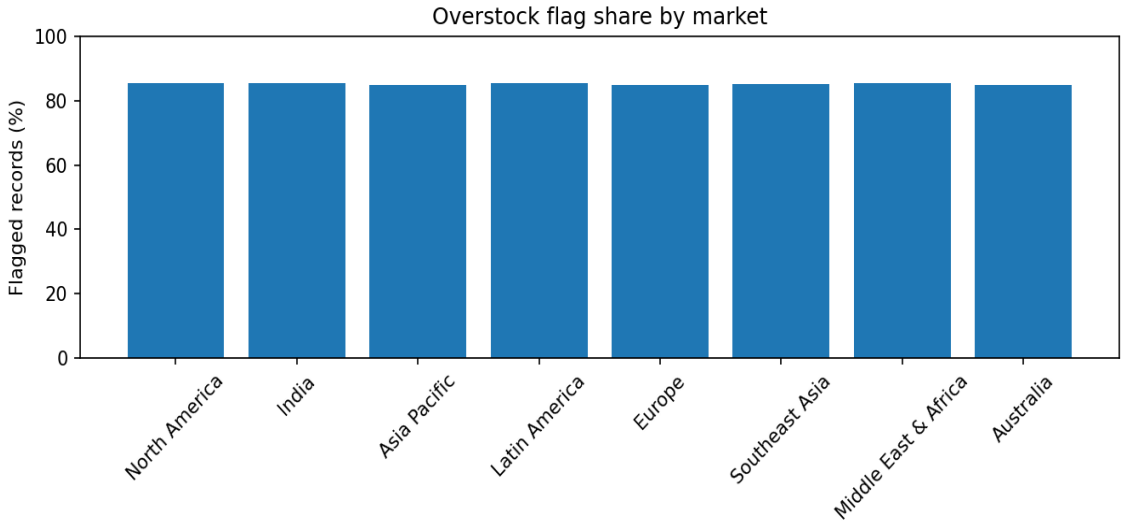
Calendar-month total comparison; differences may partly reflect uneven row counts.

Figure 3. Lead time by mode



Unadjusted average lead time by transport mode; mix and distance confound comparisons.

Figure 4. Overstock-flag share



Percentage of records flagged for overstock in each market; original flag logic unavailable.

9. Market and product findings

Market	Revenue USD	Demand units	Overstock flags
North America	1,494,425	1,716,516	5,327
India	1,371,794	1,578,718	5,272
Asia Pacific	1,329,067	1,529,720	5,305
Latin America	1,293,448	1,484,863	5,354
Europe	1,212,345	1,385,673	5,169
Southeast Asia	1,198,731	1,372,666	5,399
Middle East & Africa	1,086,575	1,247,693	5,458

Market	Revenue USD	Demand units	Overstock flags
Australia	1,005,346	1,156,238	5,314

Product	Revenue USD	Sold units
Smartwater	1,446,272	1,157,018.0
Powerade	1,369,972	1,141,643.0
Minute Maid	1,241,621	1,128,746.0
Fuze Tea	1,052,834	1,169,816.0
Coca-Cola Zero Sugar	917,098	1,146,372.0
Coca-Cola Original	876,888	1,169,184.0
Diet Coke	876,442	1,123,644.0
Sprite	800,772	1,143,960.0
Fanta	781,450	1,149,191.0
Dasani	628,382	1,142,513.0

10. Transport and cost examination

Mode	Mean lead days	Mean cost USD/row	Mean distance KM
Air	7.56	11.88	2530
Rail	7.61	9.49	2534
Truck	7.49	9.45	2515
Ship	7.44	9.40	2521

Aggregate reported transport cost = USD 478,330.34; holding = USD 699,334.92; waste = USD 33,120.34. These sums reflect the source records only, not proven actual Coca-Cola financial performance. Cost-per-shipment is not meaningful without confirming the record grain.

11. Inventory policy and optimization logic

The supplied Safety_Stock_Units and Reorder_Point_Units permit reporting of recorded thresholds and identifying rows where closing inventory is at or below the reported reorder point. No optimized safety-stock figure can be asserted without a validated demand interval, review cadence, lead-time variability and target service-level definition. Standard planning candidate: safety stock = z × standard deviation of demand during lead time; reorder point = average demand during lead time + safety stock. The formula is methodological guidance, NOT a calculated recommendation for this dataset.

Inventory turnover exists as a provided field; its underlying period, unit and accounting basis are unknown. Do not aggregate or compare as a verified financial turnover ratio until definitions and grain are confirmed.

12. Databricks and Delta Lake

A proposed notebook/script reads the same CSV, verifies 50,000 rows and 31 fields, computes market aggregations and writes Delta format to a user-authorized destination. TO BE COMPLETED: replace cloud volume placeholders, run on a Databricks workspace, verify Delta table and attach Spark/Delta evidence. No remote computation has been claimed.

13. Power BI design

The dashboard specification includes five candidate measures, market revenues, overstock share, monthly demand/forecast, transport-mode lead time, product matrix and relevant slicers. TO BE COMPLETED: create actual PBIX, validate KPI totals and capture screenshots. Be explicit that stockout flag is uniformly zero, and that 100% fill rate cannot distinguish markets.

14. Integration and verification checklist

Stage	Status / evidence needed
CSV audit	COMPLETED locally (source untouched)
Python profiling and charts	COMPLETED locally; rerunnable notebook provided
MySQL / DBeaver	TO BE COMPLETED: import, query execution, screenshots
Databricks / Delta	TO BE COMPLETED: run and capture tables / execution
Power BI	TO BE COMPLETED: build .pbix / screenshots
Data provenance	TO BE COMPLETED: original publisher/source, approval
GitHub	TO BE COMPLETED: push repository and document team members

15. Recommendations supported by these observations

1. Prioritize investigation of the large number of overstock-flagged records; confirm the flag rule, then distinguish excess from healthy buffer inventory before transferring stock. 2. Use market/product revenue and demand as descriptive allocation signals, combined with verified forecast errors, lead-time variability and service targets. 3. Audit inventory balance discrepancy before trusting reorder automation. 4. Review transport costs alongside distance and delivery mode to avoid falsely attributing cost differences to mode alone. 5. Collect credible unmet-demand or service-event data before claiming stock-out reduction impact.

16. Conclusion, references and submission gaps

The uploaded data supports reproducible descriptive assessment and a proposed end-to-end multi-tool workflow, not claims about production deployment or verified Coca-Cola corporate operations. The faculty guidance recommends an approximately 30-page final report; this is an evidence-based working report and does not pretend missing DBeaver, Databricks or Power BI screenshots have been delivered. Its final expansion should use genuine captured evidence and team-specific contributions, rather than padding.

References: (1) Supplied CocaCola_Inventory_Distribution_MySQL_Ready.csv, provenance pending; (2) Dr-Babu-BSM-PGDM-Big_Data_Project_Student_Guidelines-01.pdf, student project brief, Oct 2026. Faculty specifies PostgreSQL while this package deliberately uses MySQL per project instructions; seek faculty approval of this deviation.